

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **core banking data ingestion pipelines** for accounts, balances, transactions, postings, interest, fees, and lifecycle events using **Python** and **SQL** (PostgreSQL, MySQL, Oracle), enabling real-time visibility into **ledger integrity** and **double-entry posting sets** across **3.5 years of production support**.
- Built **idempotent streaming and batch ingestion patterns** for **event-driven banking data** using **Kafka**-equivalent architectures and **CDC (Change Data Capture)** to safely process retries and eliminate duplication in **high-throughput financial transaction systems**, achieving **99.9% reconciliation accuracy** against operational and reporting ledgers.
- Engineered **canonical domain models** and **golden source datasets** for customer, account, transaction, product, and ledger entities using **SQL schema design**, **data modelling**, and **ETL orchestration** (Python, pandas, NumPy), supporting Finance, Risk, Operations, Product, and Regulatory reporting across **SEPA/instant payments** and **cross-border payment flows**.
- Developed **event streaming pipelines** integrating core banking systems with payments, cards, KYC, fraud, CRM, and notifications platforms using **REST APIs**, **pub/sub messaging**, and **micro-batching** patterns; implemented **data lineage tracking** and **observability controls** for regulatory compliance and audit trails.
- Delivered **analytics-ready datasets** and **regulatory-grade data controls** using **Power BI**, **Tableau**, and **SQL analytics views**, supporting **GDPR/Data Protection** compliance, **ISO 20022 message flows**, and **financial services audit requirements** in a **regulated banking environment** serving **1.5M+ customers**.
- Automated end-to-end **AI-driven data pipeline orchestration** using **Claude API** (Anthropic), **Python**, **GitHub Actions CI/CD**, and **Vercel deployment** for the Ireland Jobs Tracker project, demonstrating proficiency in **prompt engineering**, **API integrations**, and **cloud-native data workflows** applicable to modern data engineering platforms.

SKILLS

Core Banking & Ledger Data Engineering – double-entry ledger design, posting sets, journals, sub-ledgers, balance reconciliation, account/transaction/product data modelling, lifecycle event streaming, idempotent ingestion patterns.

Data Platforms & Streaming – Kafka, event-driven architectures, CDC (Change Data Capture), pub/sub messaging, micro-batching, batch and streaming processing, data pipeline orchestration, high-throughput transaction processing.

SQL & Data Warehousing – PostgreSQL, MySQL, Oracle, schema design, ETL, data modelling, canonical domain models, golden source datasets, analytics-ready views, reconciliation logic, data lineage, SQL optimization.

Python Data Engineering – pandas, NumPy, scikit-learn, Python scripting for data automation, ETL development, API integrations, data validation, scheduled reporting, constraint guardrails.

Financial Domain & Regulatory Compliance – SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA/instant payments, cross-border payments, payments processing flows, open banking/PSD2, GDPR/Data Protection, audit trails, regulatory reporting, change-control management.

Analytics & Business Intelligence – Power BI (dashboards, DAX measures, data modelling), Tableau (interactive dashboards, data visualisation), Power Query for ETL, published reporting for stakeholders, business intelligence frameworks.

Cloud, DevOps & Integration – AWS (EC2, S3, Lambda, CloudWatch), Docker containerisation, Jenkins CI/CD pipelines, Git branching, GitHub Actions, REST APIs, system integrations, Linux, shell scripting, Vercel deployment, cloud-native workflows.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
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BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021
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PROJECTS

Core Banking SWIFT Payments & Ledger Operations Dashboard (HCL Technologies)

- Architected and maintained **production-grade data pipelines** for **SWIFT messaging (MT/MX) transaction flows** and core banking ledger operations, ingesting **millions of daily payment messages** into **PostgreSQL** with **idempotent processing patterns** to prevent duplication and ensure **ledger integrity**.
- Implemented **double-entry posting logic** and **real-time reconciliation workflows** between operational ledgers and reporting datasets using **SQL window functions**, **Python ETL scripts**, and **scheduled batch jobs**, achieving **100% accuracy** in balance verification for **SEPA/cross-border payment flows**.
- Built **SLA-driven operational dashboards** using **Power BI** and **Splunk monitoring** to surface **transaction-flow visibility**, **payment status tracking**, and **regulatory alerting** for banking operations teams; reduced **incident detection time by 40%** through **CloudWatch metrics** and **Dynatrace profiling**.
- Supported **GDPR/Data Protection** and **ISO 20022 message structure** compliance across data pipelines; documented **audit trails**, **change-control procedures**, and **data lineage** for regulatory review; coordinated **UAT and post-go-live support** across **interconnected ERP and payments systems** using **REST API integrations**.

Ireland Jobs Tracker – AI-Driven Data Pipeline Orchestration (Personal Project)

- Engineered end-to-end **event-driven data workflow** integrating **Apify web scrapers** for job description ingestion, **Claude API (Anthropic)** for **LLM-powered content generation**, and **reportlab PDF generation**, demonstrating mastery of **API orchestration**, **prompt engineering**, and **data transformation pipelines**.
- Implemented **structured JSON constraint guardrails** and **Claude Haiku-4.5 prompt engineering** to generate resume content at scale; deployed via **GitHub Actions CI/CD** and **Vercel**, showcasing **cloud-native automation**, **API integration patterns**, and **modern data engineering workflows** directly applicable to financial services data platforms.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)